

Module-5

Uncertain Knowledge and Reasoning

Quantifying Uncertainty- Bayes Rule -
Bayesian Belief Network- Approximate
Inference in Bayesian networks

Why do we need to quantify uncertainty?

- The world is inherently unpredictable and incomplete.
- Agents must make decisions with partial information.
- Examples:
 - **Medical diagnosis:** A patient has a symptom, what is the probability they have a disease?
 - **Weather forecasting:** What is the likelihood of rain tomorrow?
 - **Autonomous driving:** Is that a pedestrian or a shadow?

Representing Uncertainty

- We use **probability theory** to reason about uncertainty.
- Probability quantifies the **likelihood of events**.

Quantifying Uncertainty with Probability

- **Random Variable:** A variable whose value is subject to variations due to chance.

e.g., Rain = {true, false}

- **Probability Distribution, $P(X)$:** A function that describes the possible values of a random variable and their corresponding probabilities.
- **Joint Probability, $P(X,Y)$:** The probability of two or more random variables taking on specific values simultaneously.
- **Conditional Probability, $P(X|Y)$:** The probability of variable X given that variable Y has occurred.

Bayes' Rule - The Cornerstone of Reasoning

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

P(A|B) - Posterior Probability: The updated **probability of event A after observing event B.**

This is what we want to calculate.

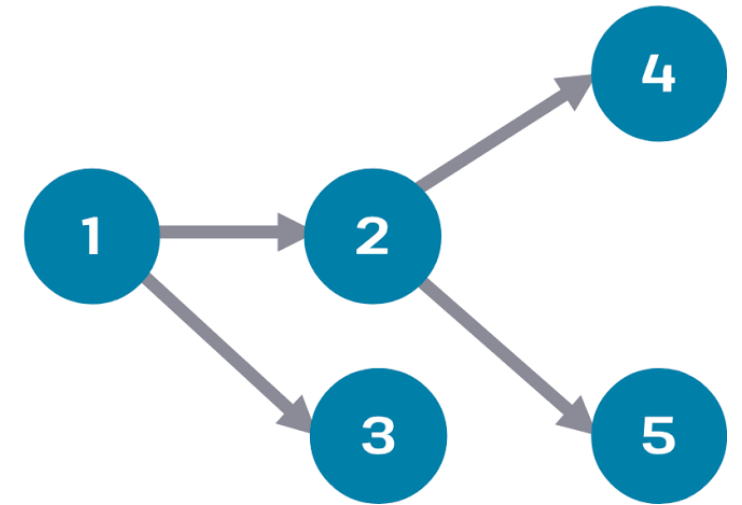
P(B|A) - Likelihood: The probability of **observing event B given that A is true.**

P(A) - Prior Probability: The initial, prior belief about the **probability of event A before observing any evidence.**

P(B) - Marginal Probability (Evidence): The total probability of observing event B.

Bayesian Belief Networks (BBNs)

- A **graphical model** that represents a set of variables and their probabilistic dependencies.
- It is a **Directed Acyclic Graph** (DAG).
- Nodes: Represent random variables (events).
- Edges: Represent direct probabilistic dependencies.
- Conditional Probability Tables (CPTs): Each node has a CPT that quantifies the effect of its parents on the node itself.



BAYESIAN NETWORKS

Introduction

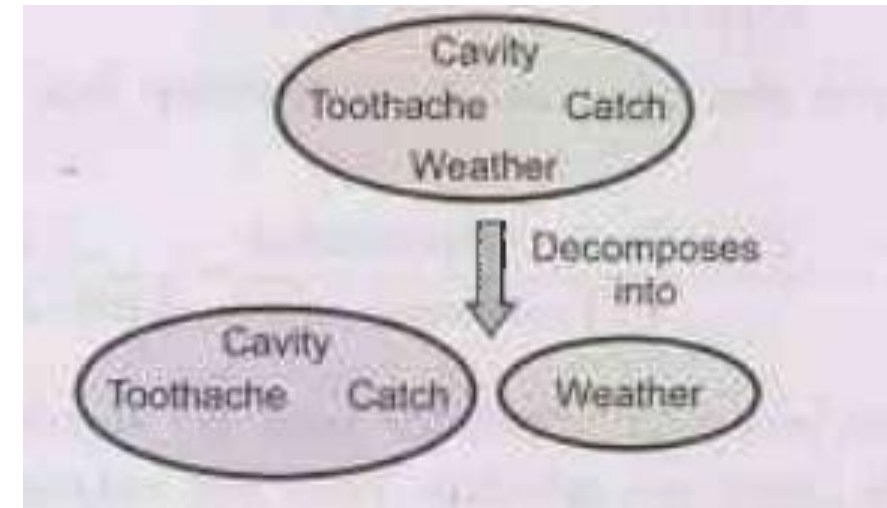
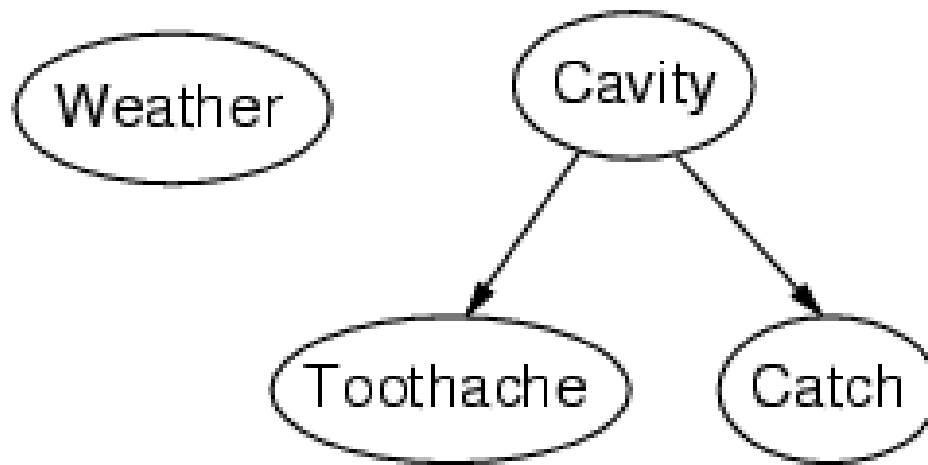
- Bayesian Networks are an explicit representation of **independence and conditional independence** relationships in simplified probabilistic form.
- *It is a directed graph in which each node is annotated with quantitative probability information.*
- Also called as:
 - Belief networks
 - Probabilistic networks
 - Causal networks
 - Knowledge map

Bayesian networks

- A simple, graphical notation for conditional independence assertions and hence for compact specification of full joint distributions.
- Syntax:
 - a set of nodes, one per variable
 - a directed, acyclic graph (link $\approx$ "directly influences")
 - a conditional distribution for each node given its parents:
$$\mathbf{P}(X_i \mid \text{Parents}(X_i))$$
- In the simplest case, the conditional distribution is represented as a **conditional probability table (CPT)** giving the distribution over X_i for each combination of parent values.

Example

- Topology of network encodes **conditional independence** assertions:



- *Weather* is independent of the other variables
- *Toothache* and *Catch* are conditionally independent given *Cavity*

Bayesian Networks

- A Bayesian network specifies a **joint distribution in a structured form**
- **Represent dependence/independence via a directed graph**
 - Nodes = random variables
 - Edges = direct dependence
- Structure of the graph $\Leftrightarrow$ Conditional independence relations

In general,

$$p(X_1, X_2, \dots, X_N) = \prod p(X_i \mid \text{parents}(X_i))$$

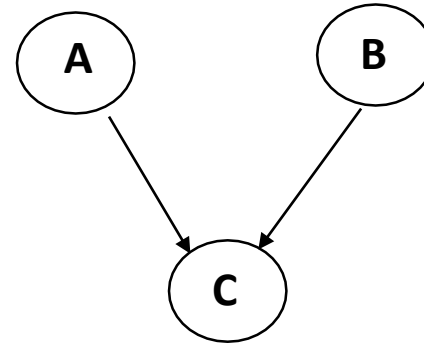
↗
The full joint distribution

↖
The graph-structured approximation

- Requires that **graph is acyclic (no directed cycles)**
- components to a Bayesian network
 - **The graph structure (conditional independence assumptions)**
 - The numerical probabilities (for each variable given its parents)

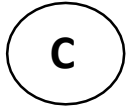
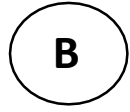
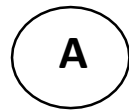
Example of a simple Bayesian network

$$p(A,B,C) = p(C|A,B)p(A)p(B)$$



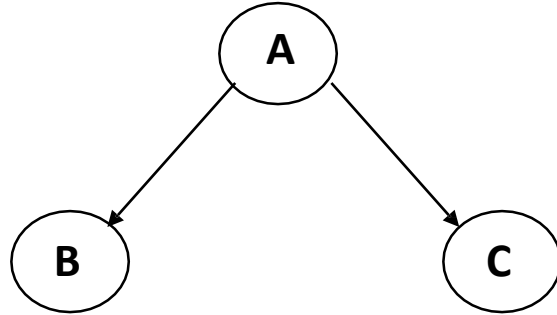
- Probability model has a simple factored form
- Directed edges => direct dependence
- Absence of an edge => conditional independence
- Also known as **belief networks, graphical models, causal networks**
- Other formulations, e.g., undirected graphical models

Examples of 3-way Bayesian Networks



Marginal Independence:
 $p(A,B,C) = p(A) p(B) p(C)$

Examples of 3-way Bayesian Networks



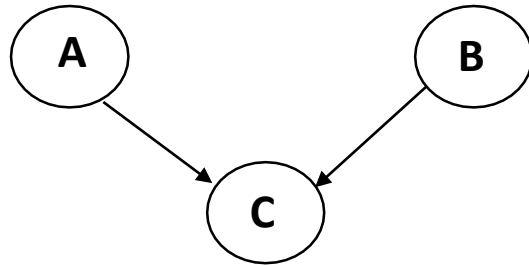
Conditionally independent effects:

$$p(A,B,C) = p(B|A)p(C|A)p(A)$$

B and C are conditionally independent
Given A

e.g., A is a **disease**, and we model
B and C as **conditionally independent
symptoms** given A

Examples of 3-way Bayesian Networks



Independent Causes:

$$p(A,B,C) = p(C|A,B)p(A)p(B)$$

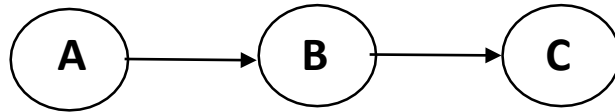
“Explaining away” effect:

Given C, **observing A makes B less likely**

e.g., earthquake/burglary/alarm example

A and B are (marginally) independent, but become dependent once C is known

Examples of 3-way Bayesian Networks



Markov dependence:
 $p(A,B,C) = p(C|B) p(B|A)p(A)$

BBNs - The "Burglary Alarm" Example

A Classic Example

•**Scenario:** You have a home alarm system. It is sometimes triggered by a burglary and sometimes by an earthquake. Your neighbors, John and Mary, will call you if they hear the alarm.

•**Variables (Nodes):**

- Burglary (B)
- Earthquake (E)
- Alarm (A)
- JohnCalls (J)
- MaryCalls (M)

Network Structure & Dependencies

- B and E are independent root nodes.
- A depends on B and E.
- J and M depend only on A.
- This network structure captures the conditional independence assumption.

Inference in Bayesian Networks

The Goal of Inference:

- To answer probabilistic queries about the variables in the network.
- Query: What is the probability of a variable given some observed evidence?
 - ✓ Example: What is the probability of a burglary given that John has called?
 - ✓ $P(B|J)$
- Evidence: The observed value of a variable.
 - ✓ Example: JohnCalls = true.
- **Inference** is the process of computing the posterior probability distribution of a set of variables, given that some other variables are observed (evidence).

Example : Burglars and Earthquakes

- You are at a party.
- Neighbour John calls to say your home alarm is ringing (but neighbour Mary doesn't).
- Sometimes your alarm is set off by minor earthquakes.
- Question: Is your home being burglarised?

Example

- Consider the following 5 binary variables:
 - B = a burglary occurs at your house
 - E = an earthquake occurs at your house
 - A = the alarm goes off
 - J = John calls to report the alarm
 - M = Mary calls to report the alarm
- We can use the full joint distribution to answer this question
 - Requires $2^5 = 32$ probabilities
 - **Can we use prior domain knowledge to come up with a Bayesian network that requires fewer probabilities?**

Constructing a Bayesian Network: Step 1

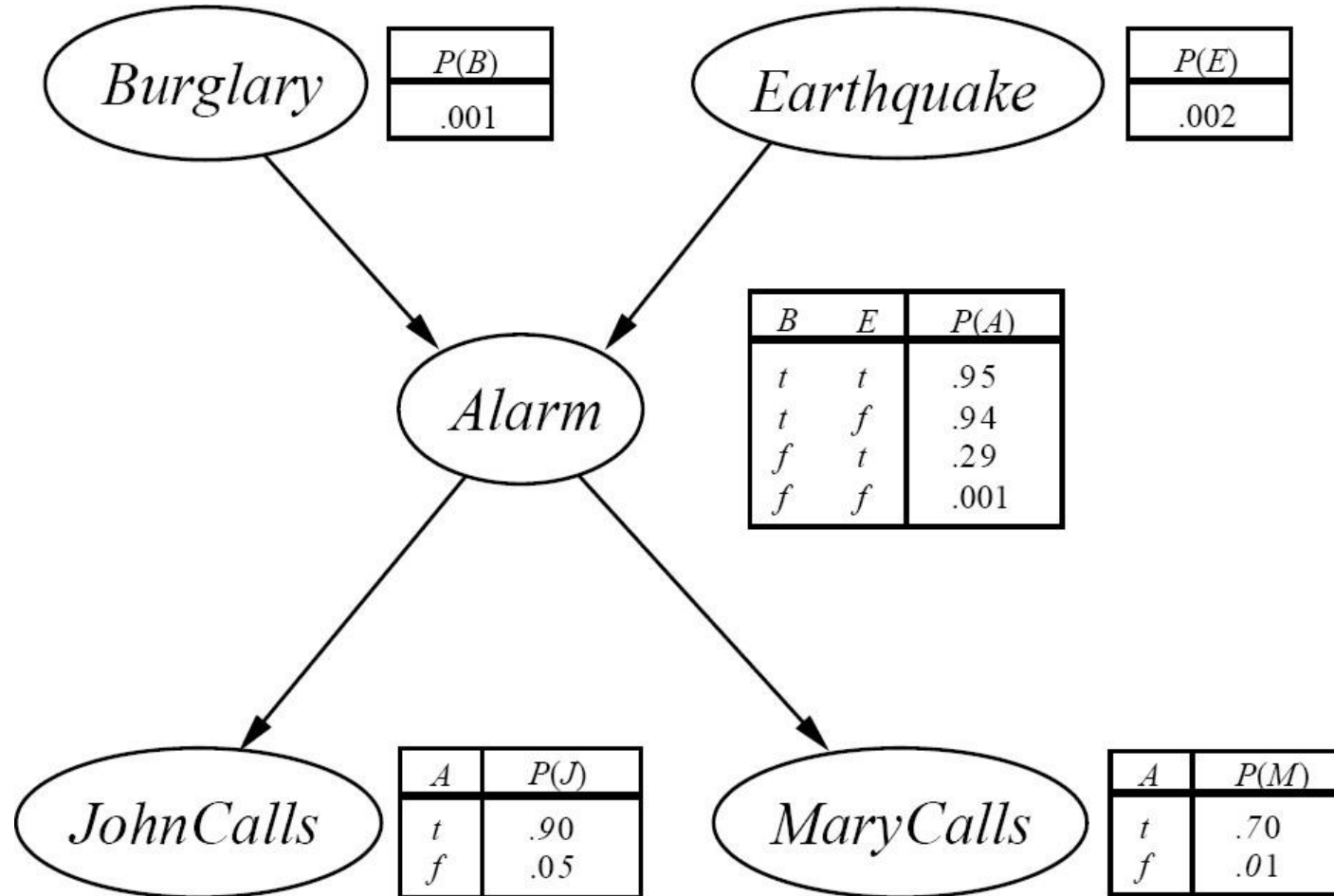
- Order the variables in terms of causality (may be a partial order)

e.g., $\{E, B\} \rightarrow \{A\} \rightarrow \{J, M\}$

- $$\begin{aligned} P(J, M, A, E, B) &= P(J, M \mid A, E, B) P(A \mid E, B) P(E, B) \\ &= P(J, M \mid A) P(A \mid E, B) P(E) P(B) \\ &= P(J \mid A) P(M \mid A) P(A \mid E, B) P(E) P(B) \end{aligned}$$

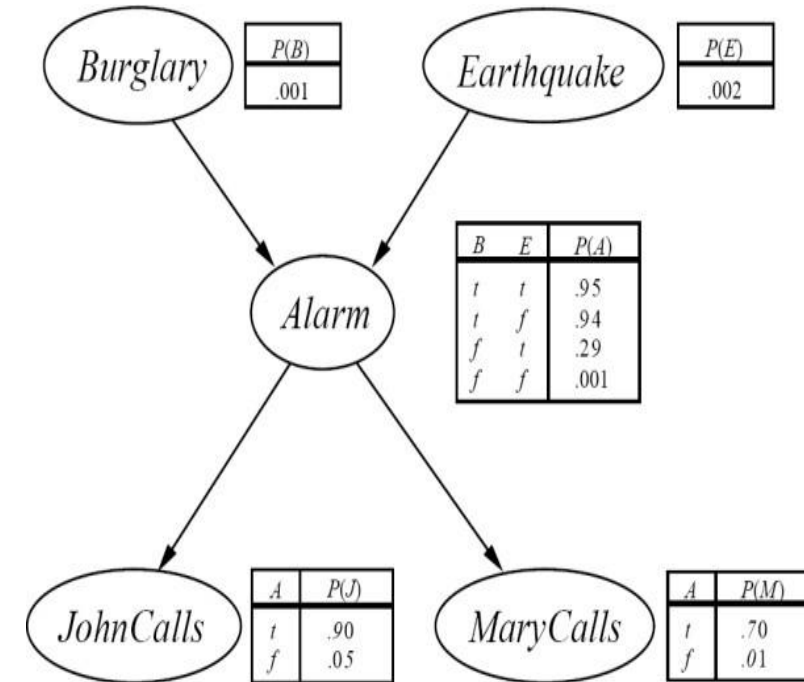
These CI assumptions are reflected in the graph structure of the Bayesian network

The Resulting Bayesian Network

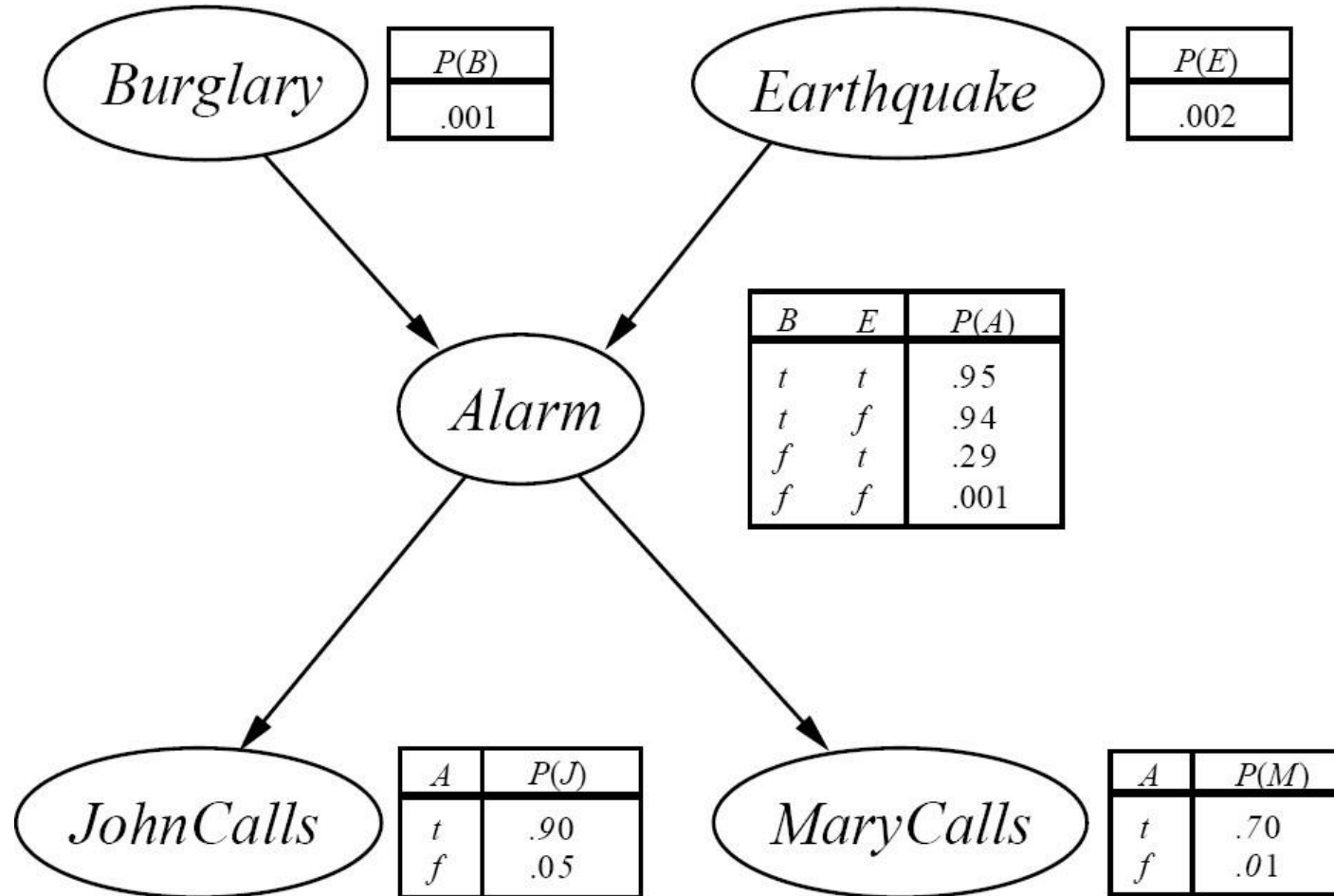


Constructing this Bayesian Network: Step 2

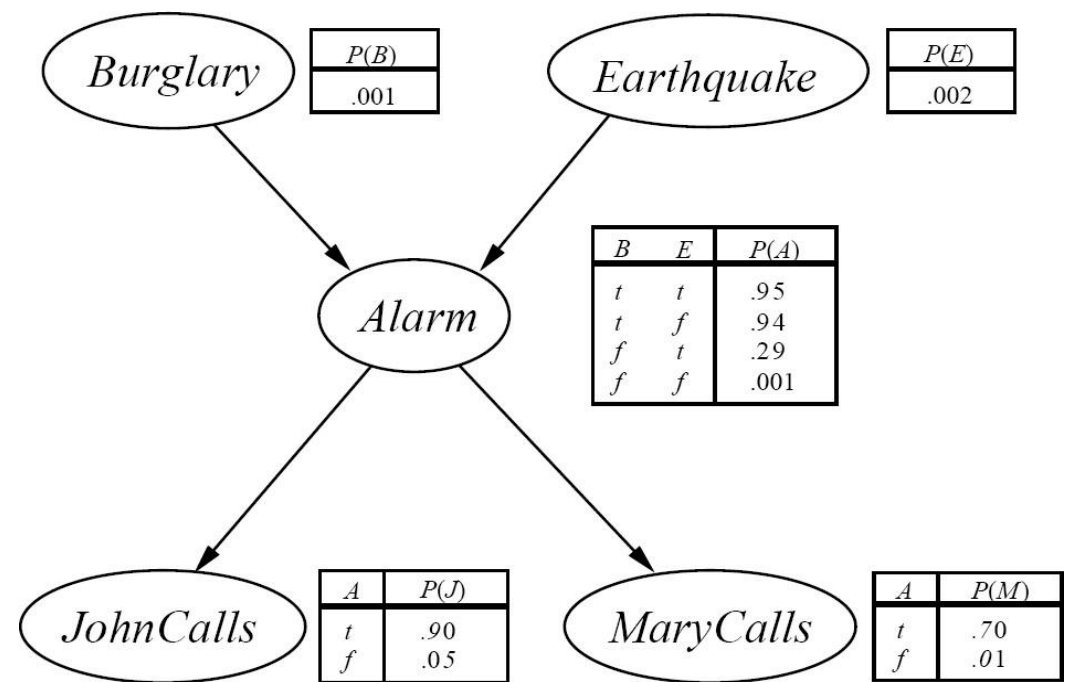
- $P(J, M, A, E, B) =$
 $P(J | A) P(M | A) P(A | E, B) P(E) P(B)$
- There are 3 conditional probability tables (CPTs) to be determined:
 $P(J | A)$, $P(M | A)$, $P(A | E, B)$
 - Requiring $2 + 2 + 4 = 8$ probabilities
- And 2 marginal probabilities $P(E)$, $P(B)$ -> 2 more probabilities
- Where do these probabilities come from?
 - Expert knowledge
 - From data (relative frequency estimates)
 - Or a combination of both



The Bayesian network



- $P(B) = 0.001$ $P(\sim B) = 0.999$
- $P(E) = 0.002$ $P(\sim E) = 0.998$
- $P(J, M, A, \sim B, \sim E) =$
- $= P(J/A) P(M/A) P(A/\sim B, \sim E) P(\sim B) P(\sim E)$
 $= 0.9 * 0.7 * 0.001 * 0.999 * 0.998$
 $= 0.00063$



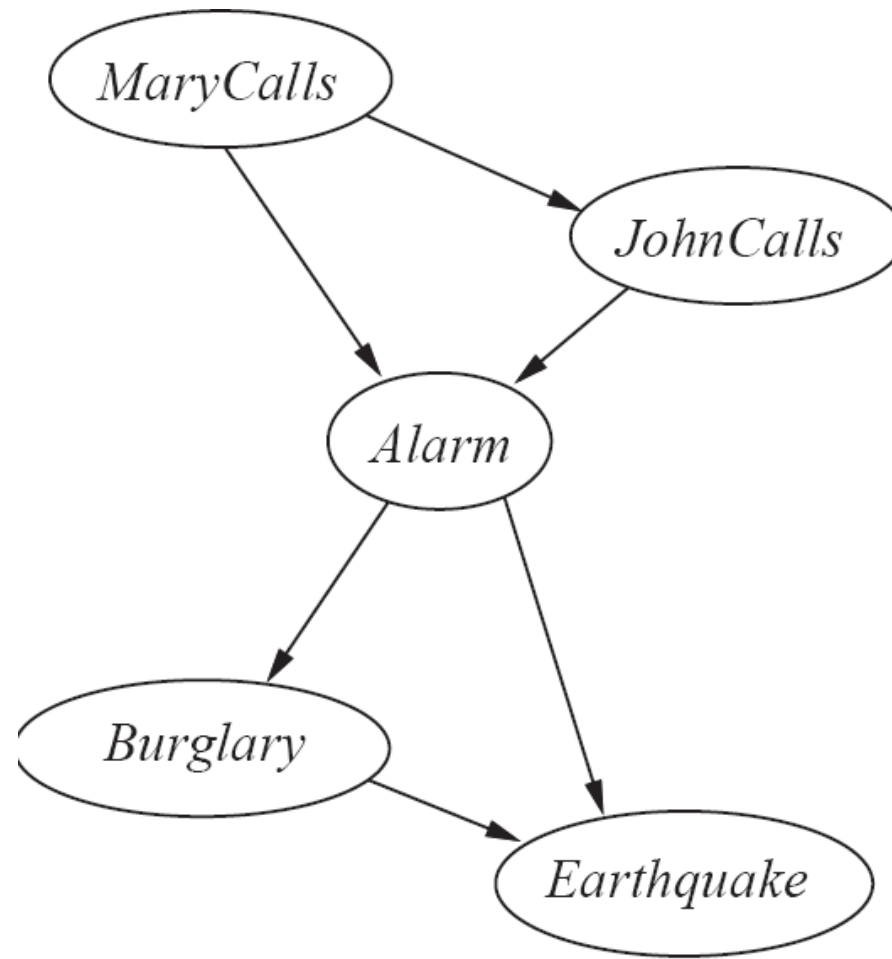
- $$P(J) = P(J/A) * P(A) + P(J/\sim A) * P(\sim A)$$

$$= 0.9 * P(A) + 0.05 * P(\sim A)$$
- $$P(A) = P(A/B, E) * P(B \wedge E) + P(A/\sim B, E) * P(\sim B \wedge E) + P(A/B, \sim E) * P(B \wedge \sim E) + P(A/\sim B, \sim E) * P(\sim B \wedge \sim E)$$

$$P(A) = 0.95 * P(B) * P(E) + 0.29 * P(\sim B) * P(E) + 0.94 * P(B) * P(\sim E) + 0.001 * P(\sim B) * P(\sim E)$$

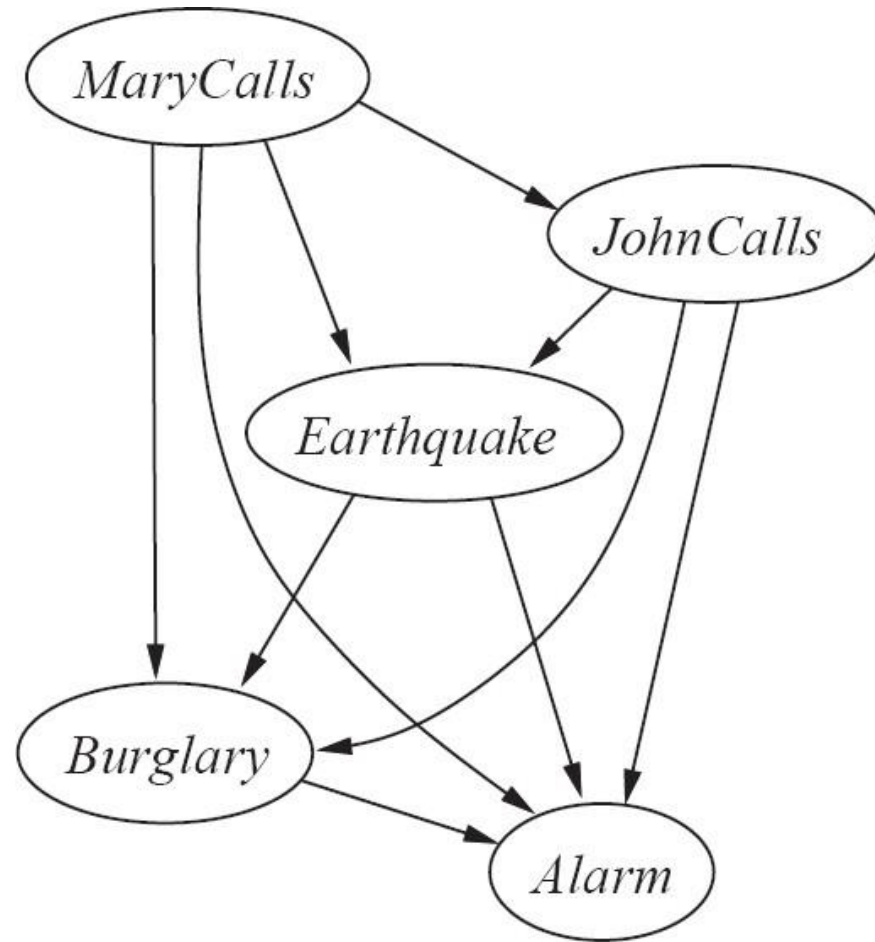
$$P(A) = 0.95 * 0.001 * 0.002 + 0.29 * 0.999 * 0.002 + 0.94 * 0.001 * 0.998 + 0.001 * 0.999 * 0.002$$
- $P(A) = 0.00252$
- $P(\sim A) = 1 - 0.00252 = 0.99748$
- $P(J) = 0.9 * 0.00252 + 0.05 * 0.99748 = 0.0022 + 0.0498 = 0.52$
- $P(J) = 0.52$

The Bayesian Network from a different Variable Ordering



(a)

The Bayesian Network from a different Variable Ordering



(b)

APPLICATIONS OF BAYESIAN NETWORKS

- Bayesian networks are used for **modelling knowledge in computational biology and bioinformatics** (gene regulatory networks, protein structure, gene expression analysis)
- **Sports betting, learning epistasis** from GWAS data sets
- **Medicine**
- **Bio-monitoring**
- **Document classification, information retrieval**
- **Semantic search**
- **Image processing, data fusion, decision support systems**
- **Engineering, gaming, law, and risk analysis.**
- There are texts applying Bayesian networks to **bioinformatics, financial and marketing informatics.**

[Genome-Wide Association Study](#)

Summary

- Bayesian networks represent a **joint distribution using a graph**
- The graph encodes **a set of conditional independence assumptions**
- Answering queries (or inference or reasoning) in a Bayesian network amounts **to efficient computation of appropriate conditional probabilities**
- Probabilistic inference is intractable in the general case
 - But it can be carried out in linear time for certain classes of Bayesian networks